

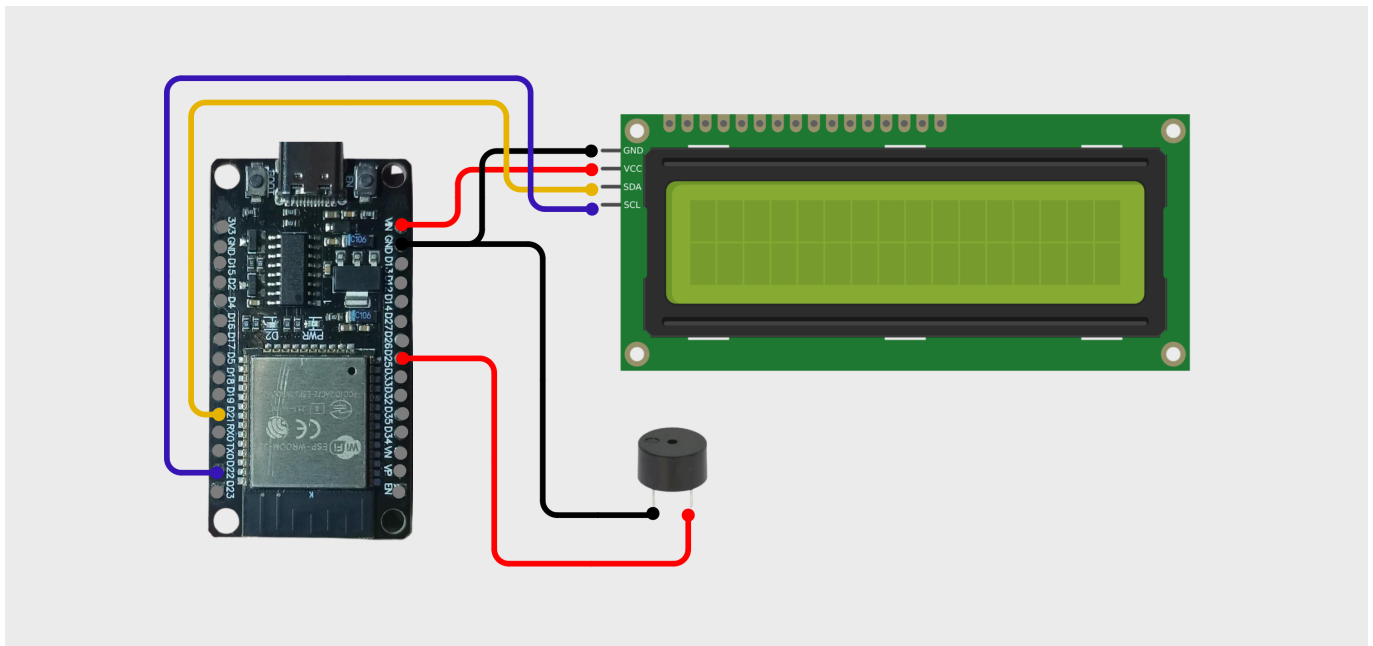
ESP32 Pomodoro Timer Fa3

A multi-mode productivity timer on ESP32. Combines an LCD readout with a web dashboard you can open from any phone on the same Wi-Fi.

Three modes:

- **Pomodoro** — automatic 25 / 5 / 15 minute rotation with a long break every 4th cycle.
- **Custom Timer** — set any countdown up to 999 minutes 59 seconds.
- **Stopwatch** — counts up from zero.

A buzzer plays a short whistle when a session ends.



Hardware

- ESP32 Dev Board
- 16x2 I2C LCD (address 0x27)
- Buzzer

Wiring

Signal	GPIO
I2C SDA	D21
I2C SCL	D22
I2C VCC	VIN
I2C GND	GND
Buzzer VCC	D25
Buzzer GND	GND

Build & Upload

```
pio run -t upload
pio device monitor -b 115200
```

After Wi-Fi connects, the serial monitor prints the IP address. Open <http://<ip>/> in any browser on the same network.

Configuration

Edit the top of `src/main.cpp`:

```
#include <Arduino.h>
#include <WiFi.h>
#include <WebServer.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// =====
// Wi-Fi
// =====

const char* WIFI_SSID = "YOUR-WIFI-SSID";
const char* WIFI_PASSWORD = "YOUR-WIFI-PASSWORD";

// =====
// Pins
// =====

#define I2C_SDA 21
#define I2C_SCL 22
#define BUZZER_PIN 25

// =====
// LCD
// =====

LiquidCrystal_I2C lcd(0x27, 16, 2);

// =====
// Web Server
// =====

WebServer server(80);

// =====
// Pomodoro Settings
// =====

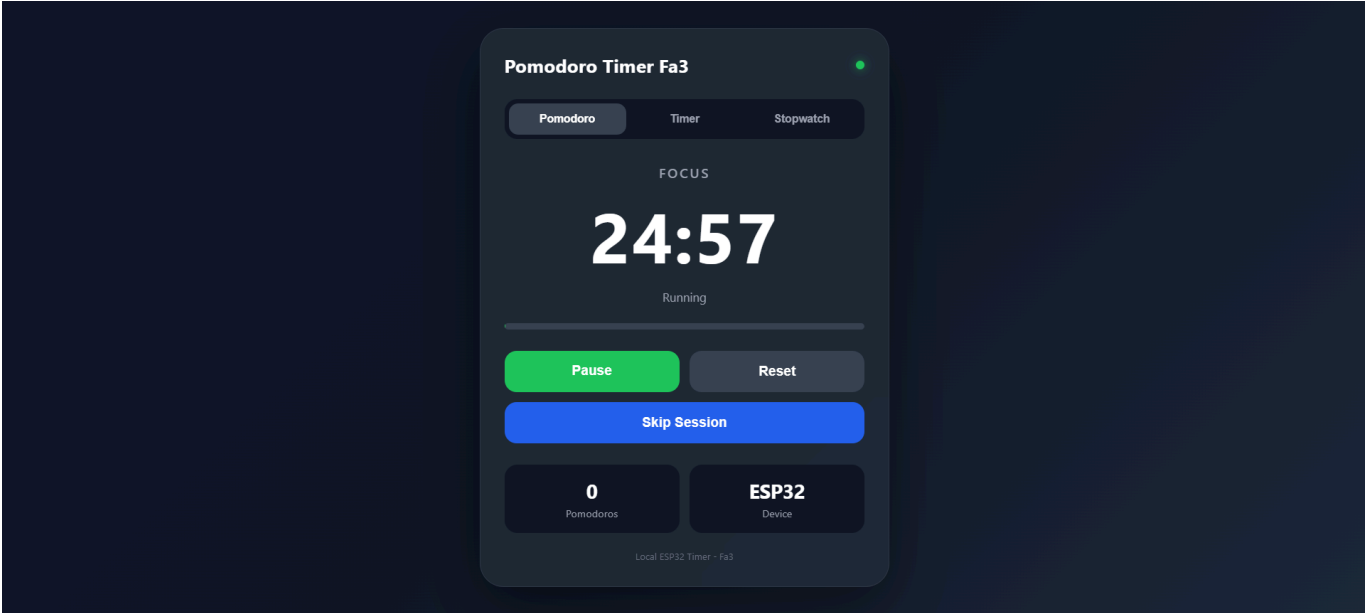
const unsigned long FOCUS_SECONDS = 25UL * 60UL;
```

```
const unsigned long SHORT_BREAK_SECONDS = 5UL * 60UL;
const unsigned long LONG_BREAK_SECONDS = 15UL * 60UL;
```

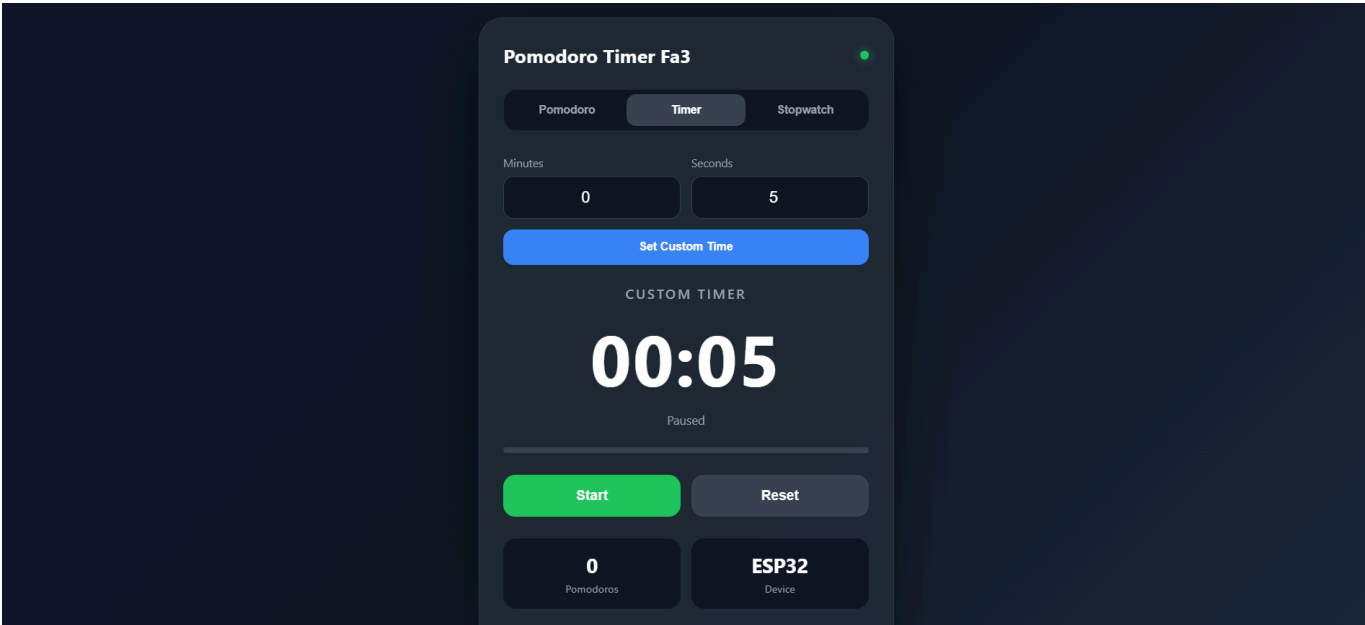
Screenshots

Web Dashboard

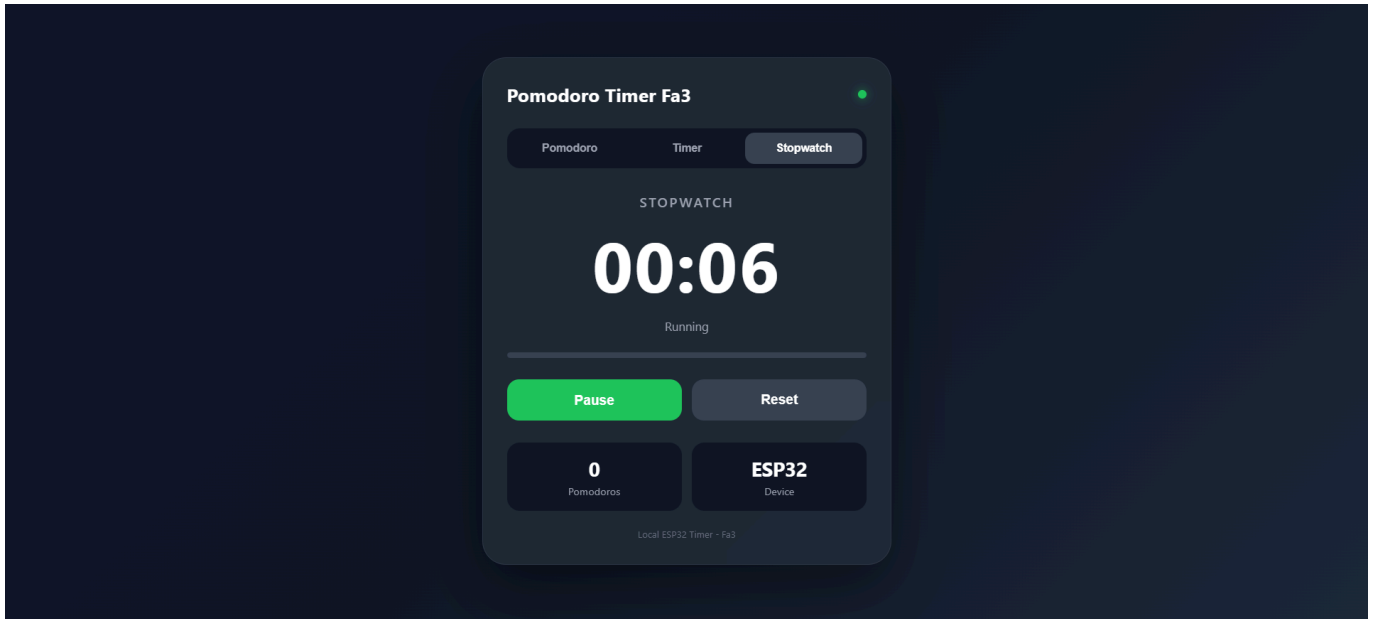
Pomodoro mode:



Custom Timer mode:

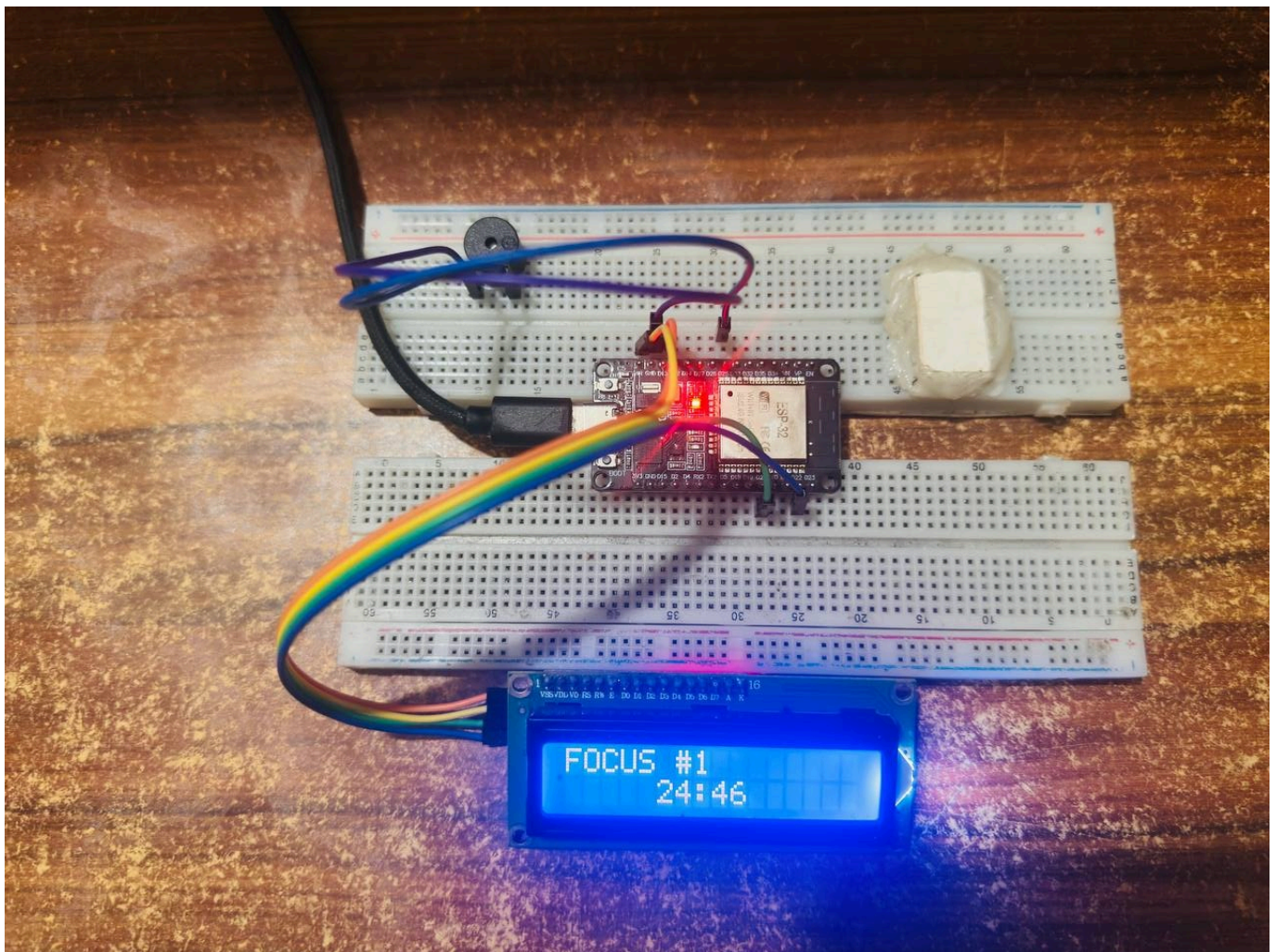


Stopwatch mode:

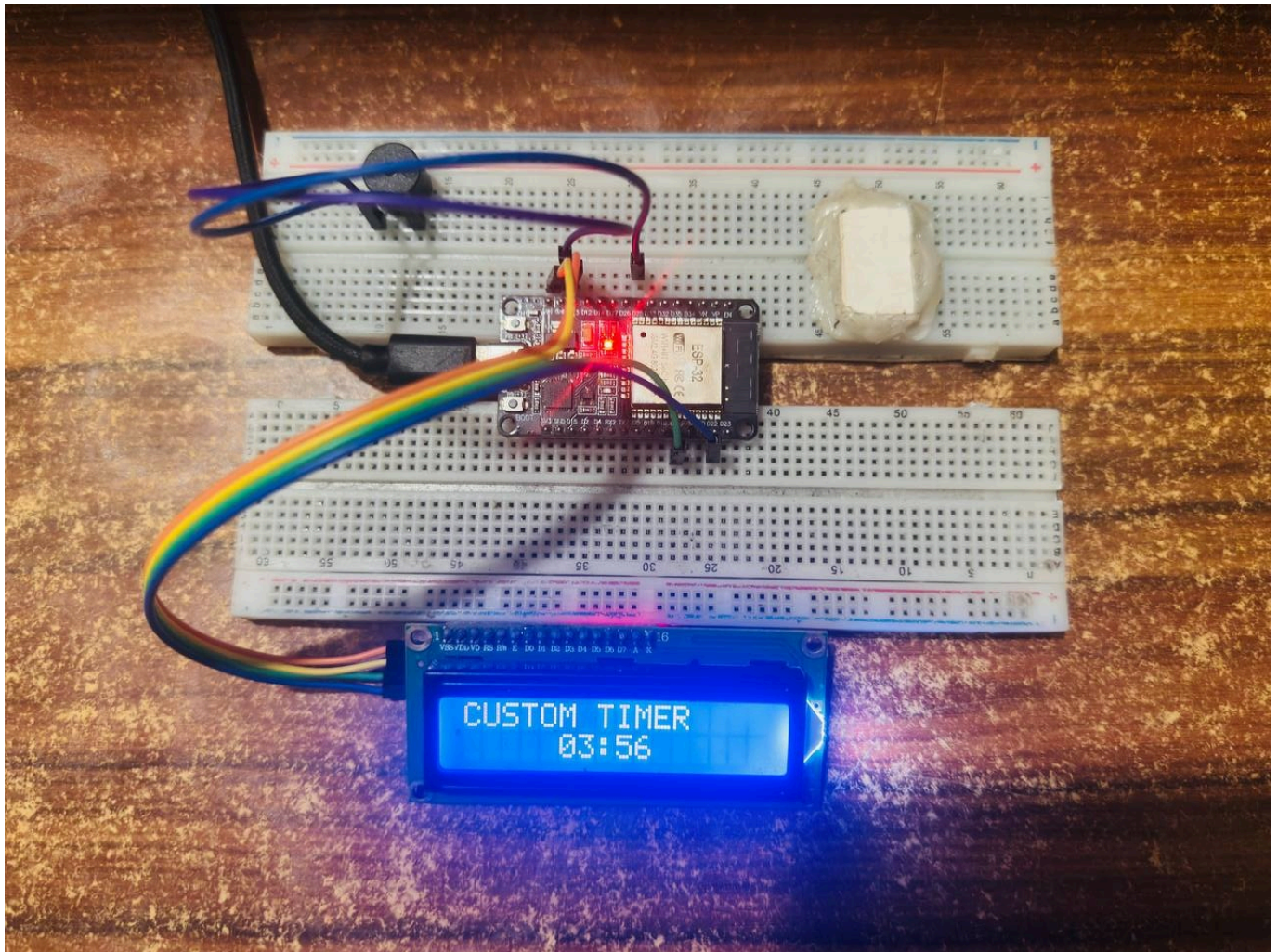


Real Device

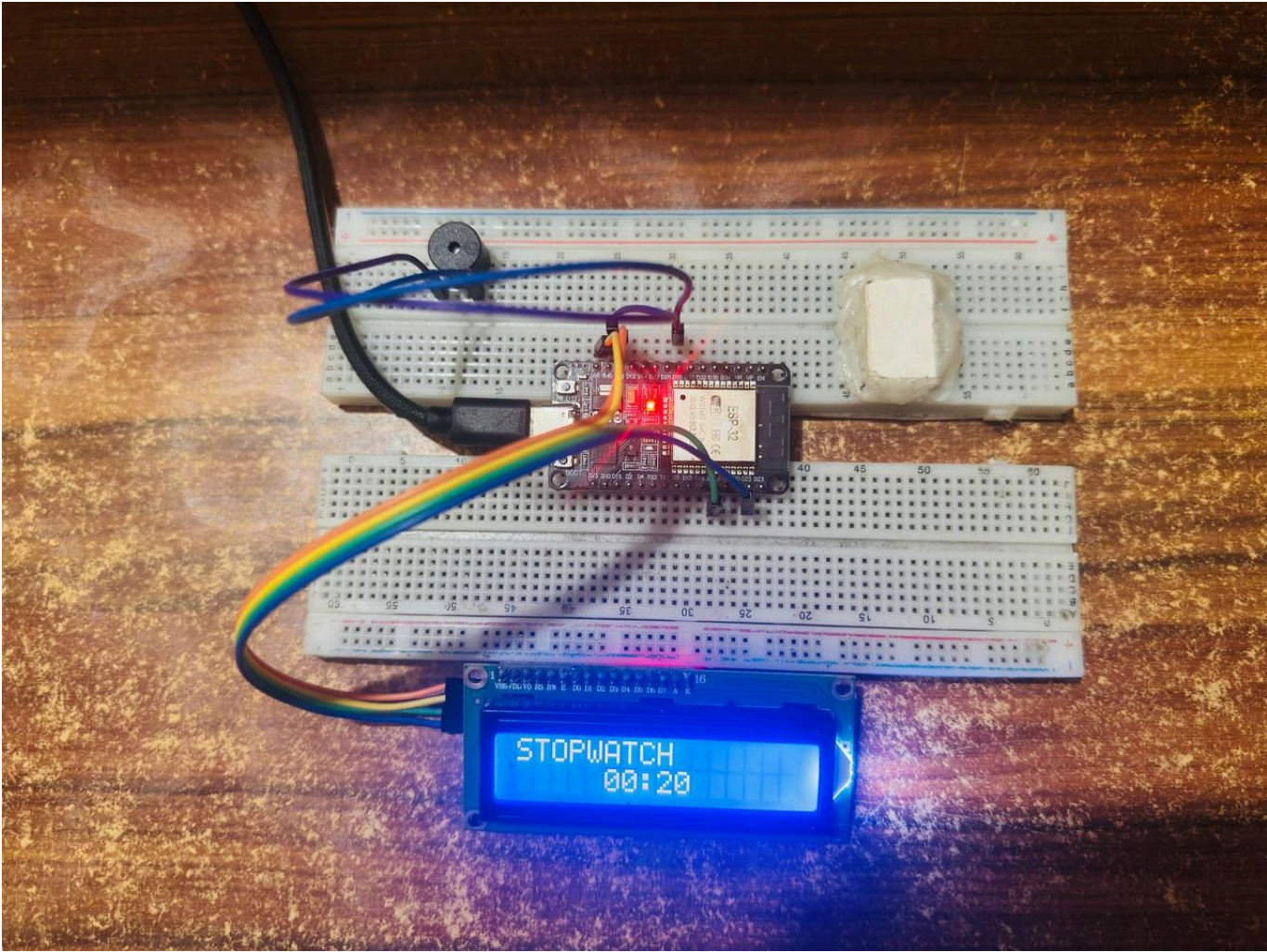
Pomodoro:



Custom Timer:



Stopwatch:



Web API

Path	Purpose
GET /	Dashboard HTML
GET /status	JSON snapshot
GET /toggle	Start/Pause
GET /reset	Reset current mode
GET /skip	Skip Pomodoro phase
GET /mode?value=pomodoro custom stopwatch	Switch mode
GET /custom?minutes=N&seconds=N	Set custom timer

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